

CADSmith — Exhibition Overview

One-page summary of the exhibit. Yellow cells are for the presenter to fill in.

Item	Entry	Notes for the organiser
Exhibit title	CADSmith — describe a part in words, get manufacturable CAD	Short form: “Text to manufacturable CAD”.
One-line pitch	Type a part in plain English or Japanese; five AI agents plan it, write CAD code and build it in a real CAD kernel; the kernel measures the result and rejects it if the dimensions are wrong.	The point of difference is the measurement, not the generation.
Category	AI / Design automation / Manufacturing software	
What a visitor sees	A live web app. A part appears in 3D in seconds, with its CadQuery source, its measured dimensions, an A3 engineering drawing and downloadable STEP / STL / DXF.	Everything on screen is produced live; nothing is a video or a mock-up.
Why it matters	Text-to-CAD systems usually generate a shape that looks right. A part that looks right but is 3 mm out cannot be manufactured. CADSmith closes the loop with exact kernel measurements, so a wrong part is caught and corrected rather than shipped.	This is the sentence to lead with for a non-technical visitor.
Who should see it	Mechanical and manufacturing engineers, CAD/PLM teams, design-automation and R&D groups, anyone evaluating generative AI for physical products.	
Time per visitor	60 seconds for the core demo; 5 minutes for the full walkthrough.	Scripts on the ‘Visitor Experience’ sheet.
Languages on the stand	English and Japanese — the entire interface, the run log, the verdicts, the drawing title block and the example prompts.	Switch in the header; a Japanese request for a standard part is understood as typed.
Interaction	Visitors may type their own part. A one-click prompt list is on screen for those who would rather watch.	
Runs without internet	Yes, in part. Standard parts (18 families), parameter edits, drawings, exports and recorded run replays all work with no network and no API key. Only the five agents need a model backend.	This is the fallback if the venue network fails.
Hardware brought	One laptop (Python 3.11, no GPU required) and, if the stand provides one, an external monitor or large display.	See the ‘Demo Setup’ sheet for the full list.
Model backend	Claude on Amazon Bedrock (Sonnet 5 for planning and code, Opus 5 as the vision Judge). A local Qwen3-VL-8B was used for the offline evaluation.	Any OpenAI-compatible endpoint also works.
Status	Working software. 33 automated tests and 11 browser test suites pass; evaluated end to end on 32 prompts against a local model.	Numbers and their sources are on the ‘Facts & Figures’ sheet.
Presenter		Name, title, organisation — fill in.
Contact		Email or QR destination — fill in.
Attribution	The research pipeline (autofab/) and its benchmark are by Jesse Barkley, Rumi Loghmani and Amir Barati Farimani (Carnegie Mellon University), arXiv:2603.26512. The exhibited application is built around that pipeline, which is unmodified.	This credit must appear on the stand and in the deck.

Every figure quoted on this sheet is sourced on the ‘Facts & Figures’ tab.

Visitor Experience — what happens, minute by minute

Two scripts: a 60-second version for a passing visitor, and a 5-minute version for someone who stays.

Time	What the presenter does	What the visitor sees	The point being made	If it goes wrong
SHORT DEMO — 60 seconds				
0:00	"Tell me a part you need." Type it, or use the visitor's own words.	An empty stand and a text box — no CAD interface to learn.	The input is language, not a modelling tool.	
0:05	Type: spur gear 50 mm diameter. Press Generate.	The gear appears in about 2 seconds, fully formed.	A standard part is served from the catalogue, exactly, with no model call at all.	Nothing to go wrong — no network is used.
0:15	Point at the Properties panel.	50 × 49.51 × 10 mm, 14,313 mm³, 129 faces, watertight solid, badged CATALOGUE.	These are the CAD kernel's own measurements, not an estimate.	
0:25	Drag the 'Teeth number' slider from 18 to 34.	The gear rebuilds with 34 real involute teeth and the size updates.	The output is parametric source, not a downloaded mesh — every number stays editable.	
0:40	Switch to the Drawing tab.	A dimensioned A3 sheet: four views, title block, first-angle projection.	It ends in something a workshop can accept — DXF and STEP download from here.	
0:55	Click 日本語.	The entire interface, log and drawing title block turn Japanese.	Built for a Japanese shop floor, not translated afterwards.	
FULL DEMO — 5 minutes				
1:00	Type a part no standard covers, e.g. a NEMA 23 motor mounting plate, 100 mm square and 8 mm thick, with a 38.1 mm pilot bore.	A design plan appears, then the agents' reasoning streams in as they work.	Five agents: Planner, Coder, Executor, Validator (vision Judge), Refiner.	If the backend is unreachable, replay a recorded run — real geometry, real verdicts.
2:00	Open the Code tab while it runs.	The CadQuery Python the Coder actually wrote, updating per iteration.	No black box: the artefact is readable, versioned source.	
2:40	Open the Validation panel.	Each claim the plan made, measured against the built solid, pass or fail.	The kernel overrules the vision Judge: a four-hole plate with one hole is blocked however confident the Judge was.	
3:30	Show the iteration filmstrip.	Every attempt kept, with its verdict — including the rejected ones.	The correction loop is visible, and failures are not hidden.	
4:10	Open the token strip.	Tokens spent per agent, and a hard ceiling per run.	A loop that will not converge is a bill; the ceiling stops it.	
4:40	Export STEP and hand over the QR / contact card.	A real STEP file lands in the downloads folder.	It leaves the stand as a file their own CAD system opens.	

Demo Setup — what the stand needs

Set Status to Ready / Pending / Blocked. The counters below read the Status column.

Area	Item	Detail	Status	Owner / note
Hardware	Laptop	Python 3.11, 8 GB RAM, no GPU. Everything runs locally except the model calls.	Pending	Presenter
Hardware	Display	External monitor or the stand's screen, 1920×1080 or better. The app is designed for a wide window.	Pending	Venue
Hardware	Power	One socket for the laptop, one for the display.	Pending	Venue
Hardware	Mouse	A real mouse — the 3D view is orbited by dragging.	Pending	Presenter
Software	Application	git checkout + python -m venv .venv + pip install -r app/requirements-app.txt	Pending	Presenter
Software	Preflight	python -m app.tools.doctor — reports Python, CadQuery, VTK, the model backend and the certificate store before the first visitor.	Pending	Presenter
Software	Headless render	On Linux install libosmesa6, or the vision Judge silently loses its renders.	Pending	Presenter
Software	Seeded runs	python -m app.tools.seed_demo_run — two runs with real geometry to replay if the network fails.	Pending	Presenter
Backend	Amazon Bedrock	Region and model access enabled for Claude Sonnet 5 and Opus 5; credentials in .env, never in the repository.	Pending	Presenter
Backend	Token ceiling	CADSMITH_TOKEN_BUDGET (250,000 per run by default) so a non-converging demo cannot run up a bill.	Pending	Presenter
Backend	Offline path	Catalogue parts, parameter edits, drawings, exports and replays need no backend at all.	Ready	Built in
Network	Outbound HTTPS	To the Bedrock endpoint. Venue Wi-Fi is enough; a phone hotspot is the backup.	Pending	Venue
Network	No inbound	The app serves on 127.0.0.1 only. Nothing is exposed.	Ready	Built in
Space	Table	Enough depth for a laptop plus the visitor to type on it.	Pending	Venue
Space	Signage	The one-line pitch, plus the Carnegie Mellon attribution.	Pending	Presenter
Content	Prompt list	On-screen benchmark prompts in both languages, one click each.	Ready	Built in
Content	Handout	QR to the repository or contact card.	Pending	Presenter
Items total			17	
Ready			3	
Still to do			14	
Percent ready			17.6%	

Risks & Fallbacks

Likelihood and Impact are 1–5. Severity multiplies them; sort by it before the show.

Risk	What it looks like on the stand	Likelihood	Impact	Severity	Mitigation already in place	Fallback on the day
Venue network is down or filtered	The five agents cannot reach the model backend.	4	4	16	Standard parts, parameter edits, drawings, exports and run replays were built to work with no backend at all.	Demo the catalogue path and replay a seeded run; both are real geometry.
Corporate TLS interception	Model calls fail with CERTIFICATE_VERIFY_FAILED.	3	4	12	The app uses the operating system's certificate store (truststore) and honours SSL_CERT_FILE / REQUESTS_CA_BUNDLE. Verification is never disabled.	Phone hotspot, or run the offline path.
A live prompt fails to build	A visitor's own wording produces no valid solid.	3	3	9	Every attempt is kept and shown with its verdict; the run log says exactly which check failed.	Say so plainly, show the failed attempt, then run a known-good prompt. A visible failure is better evidence than a hidden one.
A run will not converge and keeps spending	Iteration after iteration on a metered backend.	2	4	8	A per-run token ceiling (250,000 by default) is checked before every call; the run stops and keeps what it produced.	Stop the run and switch to a catalogue part.
The demo is slow at the wrong moment	An agent run takes minutes while someone waits.	4	3	12	Catalogue parts answer in about 2 seconds; parameter edits rebuild in about a second without a model call.	Open with the 2-second gear, start the long run, and talk over it.
Headless graphics stack missing	The vision Judge silently loses its renders.	2	3	6	The preflight names the missing rasteriser, and the health chip shows it before the first run.	Install libosmesa6, or present the kernel-measured checks, which do not need it.
A visitor asks what is actually new here	Credit for the underlying research is at stake.	3	5	15	The research pipeline is credited on the stand, in the deck and in the repository, and it is unmodified.	Answer directly: the pipeline is Carnegie Mellon's; the application, the standard-parts catalogue, the measured checks, the drawings, the spend ceiling and the bilingual interface are this work.
A visitor asks for a part with no standard	Ambiguous or under-specified request.	3	2	6	The catalogue refuses anything ambiguous and hands it to the agents rather than guessing.	Let it run, and use the design plan to show what it understood.
Laptop sleeps or updates mid-show	Black screen in front of a visitor.	2	3	6	None — this is operational.	Disable sleep and automatic updates before the doors open; keep the app running.

Highest severity
Risks scoring 12 or more

16
4

Headline mitigation: the parts of the demo a visitor sees first — the catalogue, the parameter sliders, the drawing, the exports — need no network and no API key.

Facts & Figures — every number, with its source

Nothing here is an estimate. Percentages are computed by formula from the counts beside them.

Area	Measure	Value	Derived	Source / how it was measured
Scale of the work	Application code — Python (app/)	22,300		<code>wc -l</code> over <code>app/catalog</code> , <code>app/server</code> , <code>app/tests</code> , <code>app/tools</code> on 2026-09-24. Excludes generated run artefacts.
Scale of the work	Application code — browser JavaScript	3,940		<code>wc -l</code> over <code>app/web/*</code> .js.
Scale of the work	Application code — CSS and HTML	1,576		<code>app/web/style.css</code> (1,120) + <code>app/web/index.html</code> (456).
Scale of the work	Research pipeline, unmodified (autofab/)	4,704		<code>wc -l</code> over <code>autofab/*</code> .py. Not changed by this work.
Testing	Automated tests (pytest)	33		<code>python -m pytest app/tests --collect-only</code> , 2026-09-24. 18 test modules.
Testing	Browser test suites (Playwright)	11		<code>app/tests/ui_*.py</code> — each drives the real interface in Chromium.
Catalogue	Standard part families served without a model call	18		<code>app/catalog/parts.py</code> BUILDERS: bearings, bolts, nuts, washers, screws, pins, o-rings, springs, pulleys, gears, shafts, keys, bushings, rings, collars, couplings, set screws, threaded rod.
Catalogue	Catalogue prompts built correctly (evaluation)	7		7 of 7. Local evaluation run, see below.
Catalogue	Mean time to serve a catalogue part	2.2		Seconds. Mean of 7 catalogue cases in the local evaluation (2.0–2.8 s).
Catalogue	Model tokens spent on a catalogue part	0		No model call is made; the part comes from the published standard.
Local evaluation	Prompts in the sweep	32		Five tiers: catalogue, easy, medium, hard, very hard. Model: Qwen3-VL-8B running on the presenter's own machine.
Local evaluation	Prompts that reached the model	25		7 were lost to the tunnel between the app and the local model timing out (HTTP 524) — an infrastructure limit of that setup, not a result.
Local evaluation	Prompts that produced a valid solid	22	88.0%	Of the 25 that reached the model.
Local evaluation	Prompts correct on every measured check	14	56.0%	Catalogue 7/7, easy 5/7, medium 1/6, hard 0/3, very hard 1/2.
Local evaluation	Individual dimensional checks passed	95	84.8%	95 of 112 measured checks across the sweep.
Local evaluation	Mean time for an agent-built part	121		Seconds; median 110 s. 15 non-catalogue builds. A small local model on a laptop; a hosted model is faster.
Local evaluation	Model tokens across the sweep	387,302		Reported by the provider, summed over 139 model calls.
Published research	Executable output — zero-shot, single call	95%		Barkley, Loghmani & Barati Farimani, arXiv:2603.26512, Results table.
Published research	Executable output — full pipeline, no vision	99%		Same source.
Published research	Executable output — full pipeline with vision	100%		Same source.
Published research	Mean Chamfer Distance — zero-shot	28.37		Same source. Millimetres, ICP-aligned.
Published research	Mean Chamfer Distance — full pipeline with vision	74%		Same source. A 38× reduction against zero-shot.
Published research	Median volumetric IoU — full pipeline with vision	96%		Same source.
Published research	Benchmark size	100		100 hand-written prompts in three tiers; every reference script written and inspected by hand.
Cost control	Default token ceiling per run	250,000		CADSMITH_TOKEN_BUDGET. Checked before every call; the run stops and keeps what it produced.
Output	Views on the generated drawing sheet	4		Front, left, top and isometric on an A3 sheet with a title block, first-angle projection.
Output	Export formats	5		STEP, STL, GLTF, DXF and the CadQuery source itself.
Languages	Interface languages	2		English and Japanese, including the run log, verdicts, drawing title block and benchmark prompts.

The local evaluation is evidence that the loop works end to end on a small open model, not a claim about state of the art. The published table is the research team's, measured on their benchmark.

Attribution — who did what

This division must be stated on the stand, in the deck and in any handout.

Component	What it is	Whose work
autofab/ — the CADSmith pipeline	Planner, Coder, Executor, Validator with vision Judge, Refiner; the two correction loops; retrieval over the CadQuery API.	Jesse Barkley, Rumi Loghmani, Amir Barati Farimani — Carnegie Mellon University. arXiv:2603.26512. Used unmodified.
data/ — the benchmark	100 hand-written prompts in three tiers with hand-checked reference scripts.	Same authors.
Published results	Execution rate, Chamfer Distance, F1 and volumetric IoU.	Same authors, measured on their benchmark.
app/ — the web application	Browser interface, job server, live reasoning stream, iteration filmstrip, run history, exports.	This work.
Standard-parts catalogue	18 families built to published standards and served exactly, with no model call, plus the router that refuses ambiguous requests.	This work.
Measured specification checks	The built solid is measured for what the plan claimed; a measurable shortfall blocks the version whatever the vision Judge said.	This work.
Engineering drawings	A3 sheet, four projected views, dimensions, title block, DXF export.	This work.
Parametric editing	Sliders over the generated script's own parameters; the kernel rebuilds in about a second with no model call.	This work.
Spend ceiling and instrumentation	Per-run token budget, per-agent spend, full event log.	This work.
English and Japanese interface	Including Japanese parameter edits and Japanese requests for standard parts, both without a model call.	This work.
Amazon Bedrock support	Claude on Bedrock alongside the Anthropic, OpenAI-compatible and local-model paths.	This work.

Citation

Barkley, J., Loghmani, R., & Barati Farimani, A. CADSmith: Multi-Agent CAD Generation with Programmatic Geometric Validation. arXiv:2603.26512.